INTAKE-ARBITER -- FREE-COOLING DECISION REPORT

Data centres over-cool, continuously, because nobody can promise them the hours ahead. A chiller plant needs hours of notice to change mode, and a thermometer only ever reports NOW -- so the mechanical chillers keep running through hours that outside air could have cooled for free. FortyGuard closes that gap with heat intelligence 2 m above the ground, the height a ground-mounted condenser actually breathes. This agent turns that forecast into an hour-by-hour schedule carrying a calibrated safety bound.

THE SITE

Location	Amazon Web Services, VA -- station KHEF, America/New_York
Committed pair	OSM 222672860 -> 1280340217 Amazon IAD-55 / Amazon Web Services
Facade gap	173.6 m between the two halls
Weather record	42,554 real hourly records from KHEF
Plume physics	576 steady-state solves on this site's own footprints; worst intake rise 0.3307 C at 145 deg

FortyGuard data: weather and geometry are this site's own; the four measured FortyGuard level offsets and the N-26 coverage record are Ashburn's, because only Ashburn has forecast/outcome day pairs. Quote the hours for this site; quote the coverage for Ashburn.

THIS REPORT IS A SNAPSHOT, NOT THE LIVE PAGE

The interface recomputes the decision for whatever configuration a reader selects. This file was generated at build time for the ONE configuration named below, chosen because it is the configuration in which this site's agent does the most while still changing mode at least once. If the numbers on screen differ from the numbers here, the configuration differs -- compare the two lists before concluding anything else.

Day	2022-03-16 (knife_edge)
Plant limit	18.0 C
Notice required	3 h
Forecast skill	0.50 relative to persistence
Level anchor	sensor
Condenser bank	longest facade
Switch budget	1 mode changes per day
Minimum dwell	3 h
Max dew point	15.0 C (Green Grid WP#46 p.6)

WHAT THE AGENT DID ON THIS DAY

The agent ran free cooling for 4 of 24 hours with 1 mode change. The reactive incumbent took 13 hours -- 9 more -- but declared 2 unsafe hours against the agent's 0. The incumbent had to break its own switch budget 1 time to stay safe; the agent never did. 12 hours were safe but still ran chillers, because the schedule could not afford them.

Agent	4 of 24 hours free cooling, 1 mode change(s), 0 unsafe hour(s)
Incumbent	13 of 24 hours free cooling, 2 mode change(s), 2 unsafe hour(s) the incumbent broke its own switch budget 1 time(s) to stay safe 12 hour(s) passed every gate and still ran chillers, because the schedule could not afford them

EVERY HOUR, AND THE REASON FOR IT

hh	mode	safe	binding	bound	limit	actual	margin
00	mechanical	yes	switch budget	4.814	18.0	4.609	1.074
01	mechanical	yes	switch budget	4.535	18.0	3.499	1.209
02	mechanical	yes	switch budget	5.095	18.0	4.609	1.106
03	mechanical	yes	switch budget	4.264	18.0	2.949	1.064
04	mechanical	yes	switch budget	3.429	18.0	2.389	1.010

05	mechanical	yes	switch budget	3.454	18.0	1.279	0.976
06	mechanical	yes	switch budget	3.150	18.0	1.279	1.281
07	mechanical	yes	switch budget	3.985	18.0	1.279	1.588
08	mechanical	yes	switch budget	7.319	18.0	5.169	2.317
09	mechanical	yes	switch budget	10.649	18.0	9.059	2.758
10	mechanical	yes	switch budget	13.985	18.0	15.169	2.873
11	mechanical	yes	switch budget	17.726	18.0	20.868	2.236
12	mechanical	no	dry-bulb	18.861	18.0	21.431	1.703
13	mechanical	no	dry-bulb	20.796	18.0	21.428	1.350
14	mechanical	no	dry-bulb	23.216	18.0	21.968	1.205
15	mechanical	no	dry-bulb	22.668	18.0	21.428	1.127
16	mechanical	no	dry-bulb	22.148	18.0	21.428	1.210
17	mechanical	no	dry-bulb	21.498	18.0	20.311	1.364
18	mechanical	no	dry-bulb	19.868	18.0	18.026	1.661
19	mechanical	no	dry-bulb	18.295	18.0	15.317	1.803
20	FREE	yes	-	17.941	18.0	16.348	1.886
21	FREE	yes	-	16.105	18.0	14.685	1.765
22	FREE	yes	-	14.592	18.0	13.643	1.653
23	FREE	yes	-	14.685	18.0	12.537	1.299

bound = the agent's 90 %-nominal upper bound on intake air, forecast plus margin
actual = what the intake temperature turned out to be, from the station record
margin = group-conditional forecast error for that hour of day, plus plume spread

THE REASONING, HOUR BY HOUR

00:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 4.814 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

01:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 4.534 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

02:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 5.094 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

03:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 4.264 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

04:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 3.429 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

05:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 3.454 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

06:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 3.149 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

07:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 3.984 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

08:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 7.319 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

09:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 10.649 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

10:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 13.984 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

11:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 17.726 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

12:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 18.861 C against a 18.0 C limit, so it fails by 0.861 C. It would take a limit 0.861 C higher, or a bound 0.861 C tighter, to change this hour.

13:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 20.796 C against a 18.0 C limit, so it fails by 2.796 C. It would take a limit 2.796 C higher, or a bound 2.796 C tighter, to change this hour.

14:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 23.216 C against a 18.0 C limit, so it fails by 5.216 C. It would take a limit 5.216 C higher, or a bound 5.216 C tighter, to change this hour.

15:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 22.668 C against a 18.0 C limit, so it fails by 4.668 C. It would take a limit 4.668 C higher, or a bound 4.668 C tighter, to change this hour.

16:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 22.148 C against a 18.0 C limit, so it fails by 4.148 C. It would take a limit 4.148 C higher, or a bound 4.148 C tighter, to change this hour.

17:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 21.498 C against a 18.0 C limit, so it fails by 3.498 C. It would take a limit 3.498 C higher, or a bound 3.498 C tighter, to change this hour.

18:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 19.868 C against a 18.0 C limit, so it fails by 1.868 C. It would take a limit 1.868 C higher, or a bound 1.868 C tighter, to change this hour.

19:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 18.295 C against a 18.0 C limit, so it fails by 0.295 C. It would take a limit 0.295 C higher, or a bound 0.295 C tighter, to change this hour.

20:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 17.941 C, which is 0.059 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 1.886 C, and the margin is measured, not chosen: 1.726 C of group-conditional forecast error for this hour of day, plus 0.1605 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

21:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 16.104 C, which is 1.896 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 1.765 C, and the margin is measured, not chosen: 1.575 C of group-conditional forecast error for this hour of day, plus 0.1897 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

22:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 14.592 C, which is 3.408 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 1.653 C, and the margin is measured, not chosen: 1.456 C of group-conditional forecast error for this hour of day, plus 0.1970 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

23:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 14.685 C, which is 3.315 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 1.299 C, and the margin is measured, not chosen: 1.109 C of group-conditional forecast error for this hour of day, plus 0.1896 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

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